

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CHEMISTRY

9729/03

Paper 3 Free Response

13 September 2017

2 hours

Candidates answer on separate paper.

Additional Materials: Writing Papers
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of **20** printed pages.



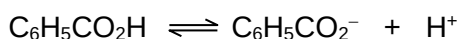
Section A

Answer **all** the questions in this section.

- 1 Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$ is a colourless crystalline solid and a simple aromatic carboxylic acid. Benzoic acid occurs naturally in many plants. Salts of benzoic acid are used as food preservatives and benzoic acid is an important precursor for the industrial synthesis of many other organic substances.

- (a) In the identification of benzoic acid, a commonly used reagent is neutral iron(III) chloride solution, in which a buff precipitate of iron(III) benzoate, $\text{Fe}(\text{C}_6\text{H}_5\text{CO}_2)_3$ is formed. When 50.0 cm^3 of iron(III) chloride solution was added to 50.0 cm^3 of benzoic acid solution, 0.0532 g of the buff precipitate was formed in the mixture.

The reactions that take place are shown below:

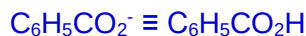


(Given: M_r of $\text{Fe}(\text{C}_6\text{H}_5\text{CO}_2)_3 = 418.8$)

- (i) Calculate the number of moles of benzoic acid that reacted with the neutral iron(III) chloride solution.

[1]

$$\begin{aligned} \text{Number of moles of } \text{Fe}(\text{C}_6\text{H}_5\text{CO}_2)_3 &= 0.0532 / 418.8 \\ &= 1.2703 \times 10^{-4} \text{ mol} \end{aligned}$$



$$\begin{aligned} \text{Number of moles of benzoic acid reacted} &= 3 \times 1.2703 \times 10^{-4} \\ &= 3.8108 \times 10^{-4} \\ &= 3.81 \times 10^{-4} \text{ mol} \end{aligned}$$

- (ii) Given that the pH of the solution after the reaction is 2.33, calculate the number of moles of H^+ in the mixture.

[1]

$$[\text{H}^+] = 10^{-2.33} = 4.677 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\begin{aligned} \text{Number of moles of } \text{H}^+ &= 0.100 \times 4.677 \times 10^{-3} \\ &= 4.677 \times 10^{-4} \\ &= 4.68 \times 10^{-4} \text{ mol} \end{aligned}$$

- (iii) Assuming that the H^+ ions in solution are formed only from the dissociation of benzoic acid in solution, as well as the reaction of benzoic acid with iron(III) chloride, calculate the concentration of benzoate ions in the mixture.

[2]

Since the H^+ comes from both dissociation of benzoic acid and reaction of benzoic acid with iron (III) chloride,

$$\begin{aligned} \text{Number of moles of benzoate in solution} &= 4.677 \times 10^{-4} - 3.8108 \times 10^{-4} \\ &= 8.665 \times 10^{-5} \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{Concentration of benzoate in solution} &= \text{moles} / \text{volume} \\ &= 8.665 \times 10^{-5} / [(50 + 50)/1000] \\ &= 8.67 \times 10^{-4} \text{ mol dm}^{-3} \end{aligned}$$

Allow ecf

- (iv) Iron(III) benzoate is a sparingly soluble salt.

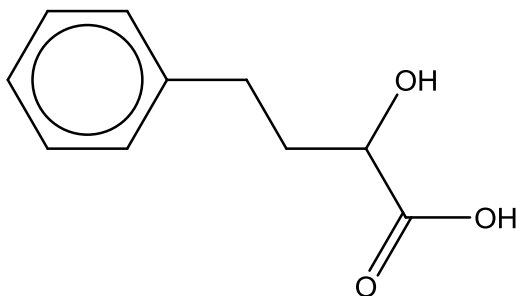
Calculate the K_{sp} of iron(III) benzoate, given that the concentration of Fe^{3+} in the mixture was $1.83 \times 10^{-3} \text{ mol dm}^{-3}$ at equilibrium.

[1]

$$\begin{aligned} K_{sp} &= [\text{Fe}^{3+}] [\text{C}_6\text{H}_5\text{CO}_2^-]^3 \\ &= (1.83 \times 10^{-3}) (8.6655 \times 10^{-4})^3 \\ &= 1.1908 \times 10^{-12} \\ &= 1.19 \times 10^{-12} \text{ mol}^4 \text{ dm}^{-12} \end{aligned}$$

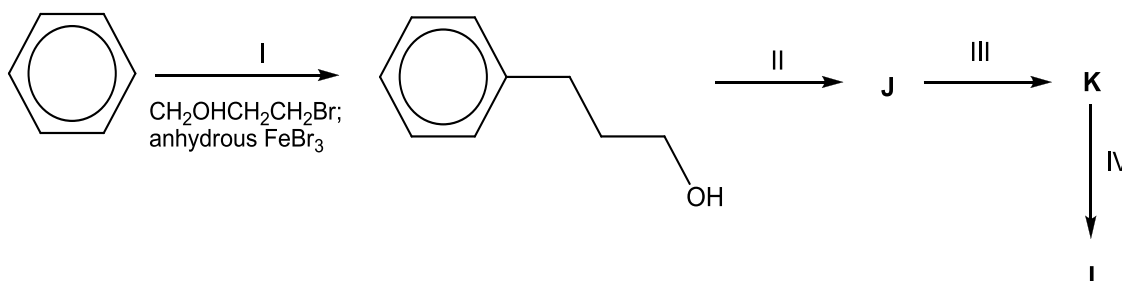
Allow ecf

(b) Benzoic acid can be produced by oxidising compound L.



Compound L

Compound L can be synthesised by the following reaction scheme.



(i) State the *type of reaction* in step I and hence explain the need for FeBr_3 to be anhydrous.

[2]

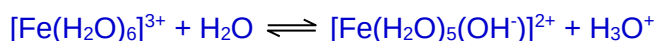
Electrophilic substitution

The catalyst will be hydrolysed and lose its catalytic property if it is dissolved in water.

FeBr_3 has to be anhydrous as it has to be electron deficient to accept a lone pair of electron from the Br in $\text{CH}_3(\text{CH}_2)_3\text{Br}$ to form the electrophile $\text{CH}_3(\text{CH}_2)_3^+$ for electrophilic substitution to take place.

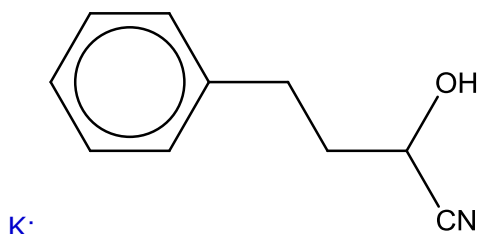
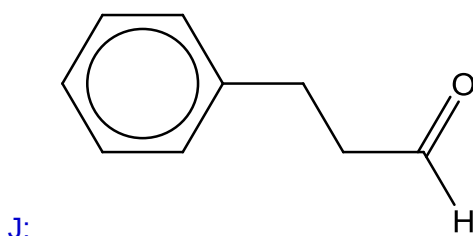
If water is present, $\text{Fe}^{3+}(\text{aq})$ will be present and the electrophile $\text{CH}_3(\text{CH}_2)_3^+$ will not be formed.

$\text{Fe}^{3+}(\text{aq})$ will undergo hydrolysis:



(ii) Suggest the identities of J and K.

[2]



- (iii) State the reagents and conditions for steps II to IV.

[3]

II: $\text{K}_2\text{Cr}_2\text{O}_7$, dilute H_2SO_4 , heat with immediate distillation

III: HCN , trace amount of NaCN or NaOH , cold

IV: dilute H_2SO_4 or dilute HCl , heat

- (iv) State the *type of reaction* for stage IV.

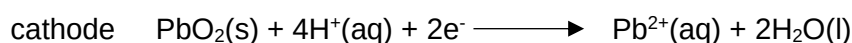
[1]

Acidic hydrolysis

[Total: 13]

- 2 Sulfur is a chemical element with the symbol S and an atomic number of 16. It is an abundant, multivalent non-metal. It can be found in amino acids and as a precursor to other chemicals such as H_2SO_4 . H_2SO_4 is a common mineral acid with many uses. It can be used as an electrolyte in batteries.

- (a) In the cells of a lead-acid car battery the following reactions take place.



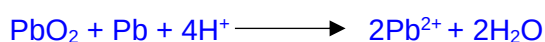
- (i) Use the *Data Booklet* to calculate $E^\ominus_{\text{cell}}$ for this reaction.

[1]

$$E^\ominus_{\text{cell}} = 1.47 - (-0.13) = +1.60\text{V}$$

- (ii) Construct an equation for the overall reaction.

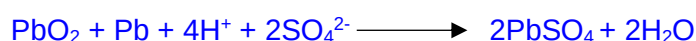
[1]



The electrolyte in a lead-acid cell is $\text{H}_2\text{SO}_4(\text{aq})$. Most of the $\text{Pb}^{2+}(\text{aq})$ ions that are produced at the electrodes are precipitated as a highly insoluble $\text{PbSO}_4(\text{s})$.

- (iii) Construct an equation for the overall cell reaction in the presence of H_2SO_4 .

[1]



- (iv) By considering the effect of decreasing $[\text{Pb}^{2+}(\text{aq})]$ on the electrode potentials of the cathode and the anode, state and explain whether the overall $E^\ominus_{\text{cell}}$ will increase, decrease or remain the same.

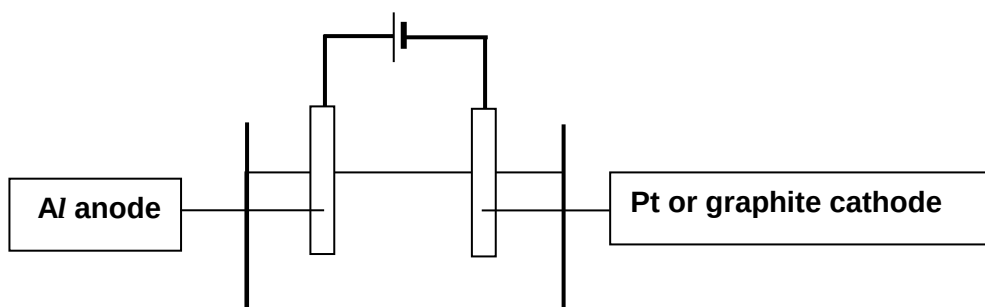
[1]

Overall $E^\ominus_{\text{cell}}$ will increase as $[\text{Pb}^{2+}]$ decreases, $E^\ominus(\text{PbO}_2/\text{Pb}^{2+})$ will become more positive, but $E^\ominus(\text{Pb}^{2+}/\text{Pb})$ will become more negative

- (b) H_2SO_4 is also used as an electrolyte in the anodising of aluminium. Anodising of aluminium is the process of coating aluminium metal with aluminium oxide (Al_2O_3) via electrolysis with the aluminium metal as the anode.

- (i) Using $\text{H}_2\text{SO}_4(\text{aq})$ as the electrolyte and an inert electrode, draw an electrolysis set-up to show how a piece of aluminium metal can be anodised.

[1]

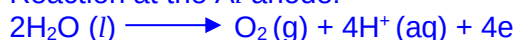


- Correct orientation of the battery
- Correct labeling of both the electrodes with the material used
- Correct cell. i.e. An electrolytic cell and not a electrochemical cell

- (ii) Write chemical equations to show the reactions at the anode and cathode during anodising. Include in your answers, the overall equation.

[3]

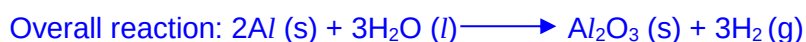
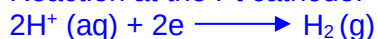
Reaction at the Al anode:



The oxygen liberated then reacts with the aluminium to produce aluminium oxide protective surface.



Reaction at the Pt cathode:



- (iii) The aluminium piece to be anodised has a surface area of 29.2 cm^2 . Calculate the time taken to form a 0.2 mm protective layer of Al_2O_3 on the aluminium piece if a current of 2.0 A is passed through the set-up.

(Density of Al_2O_3 is 3.95 g cm^{-3})

[4]

$$\begin{aligned} \text{Volume of Al}_2\text{O}_3 \text{ layer} &= 29.2 \times 0.02 \\ &= 0.58400 \text{ cm}^3 \end{aligned}$$

$$\begin{aligned} \text{Mass of Al}_2\text{O}_3 &= 3.95 \times 0.584 \\ &= 2.3068 \text{ g} \end{aligned}$$

$$\text{Amount of Al}_2\text{O}_3 = \frac{2.3068}{2 \times 27.0 + 3 \times 16.0}$$

$$\begin{aligned}
 &= 0.022616 \text{ mol} \\
 \text{Amount of O}_2 &= 0.0226 \times \frac{3}{2} \\
 &= 0.033923 \text{ mol} \\
 \text{Amount of electrons passed through the anode} &= 0.0339 \times 4 \\
 &= 0.13569 \text{ mol} \\
 Q &= 0.13569 \times 96500 \\
 &= 13094 \text{ C} \\
 \text{Time needed} &= 13094 \div 2.0 \\
 &= 6547.2 \text{ s} = 6550 \text{ s (3s.f)}
 \end{aligned}$$

Alternatively from here,
It = n_eF approach

$$\begin{aligned}
 t &= (n_e F) / I \\
 &= (0.13569 \times 96500) / 2.0 \\
 &= 6547.2 \text{ s} = 6550 \text{ s (3s.f)}
 \end{aligned}$$

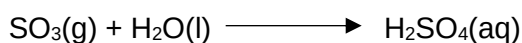
- (iv) Give one example of an anodised aluminium object, and explain the advantages of anodising it.

[1]

Examples of anodized aluminium objects: drink cans, windows frames and grilles

Anodised aluminium objects are more resistant to corrosion. Moreover, they can be decorative as well since the aluminium oxide layer is able to absorb dyes.

- (c) H₂SO₄ can be produced when sulfur trioxide is added to water, according to the following equation:



Calculate the volume of gaseous SO₃ under room temperature and pressure, needed to form 30.0 cm³ of 0.0200 mol dm⁻³ of H₂SO₄.

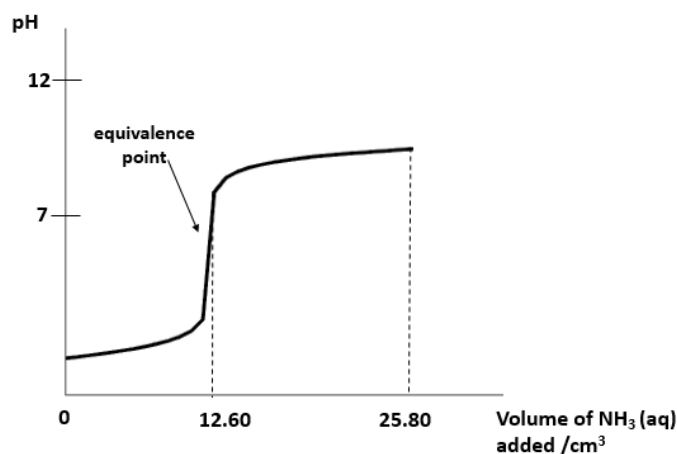
[1]

$$\text{Amount of H}_2\text{SO}_4 \text{ to be produced} = \frac{30}{1000} \times 0.02 = 6.00 \times 10^{-4}$$

$$\text{Amount of SO}_3 \text{ needed} = 6.00 \times 10^{-4}$$

$$\begin{aligned}
 \text{Volume of SO}_3 \text{ needed at r.t.p} &= 6.00 \times 10^{-4} \times 24.0 \text{ dm}^3 \\
 &= 0.0144 \text{ dm}^3
 \end{aligned}$$

- (d) 25.0 cm³ of 0.0200 mol dm⁻³ of H₂SO₄ was titrated against NH₃(aq). The following graph was obtained.



- (i) Calculate the initial pH of the sulfuric acid solution.

[1]

$$[\text{H}_2\text{SO}_4] = 0.02 \text{ mol dm}^{-3}$$

$$[\text{H}^+] = 0.04 \text{ mol dm}^{-3}$$

$$\text{pH} = 1.40$$

- (ii) Calculate the concentration of NH_3 (aq) used in this titration.

[2]



$$\text{Amount of H}_2\text{SO}_4 \text{ used in titration} = \frac{25}{1000} \times 0.02 = 5.000 \times 10^{-4}$$

$$\text{Amount of NH}_3 \text{ used in titration} = \frac{25}{1000} \times 0.02 \times 2 = 1.000 \times 10^{-3}$$

$$[\text{NH}_3] = \frac{1 \times 10^{-3}}{\frac{12.60}{1000}} = 0.0794 \text{ mol dm}^{-3}$$

- (iii) Suggest a suitable indicator for this titration, giving a reason for your choice.

[2]

Methyl orange.

The working range for methyl orange (is in the region of 3 to 5 and) coincides with the equivalence point of a strong acid-weak base titration (<7).

[5]

- (e) Thiophenol are the sulfur analogue of phenol, that is, sulfur takes the place of oxygen in the hydroxyl group of phenol. A common reaction that phenol and thiophenol have is shown in **Figure 1** below.

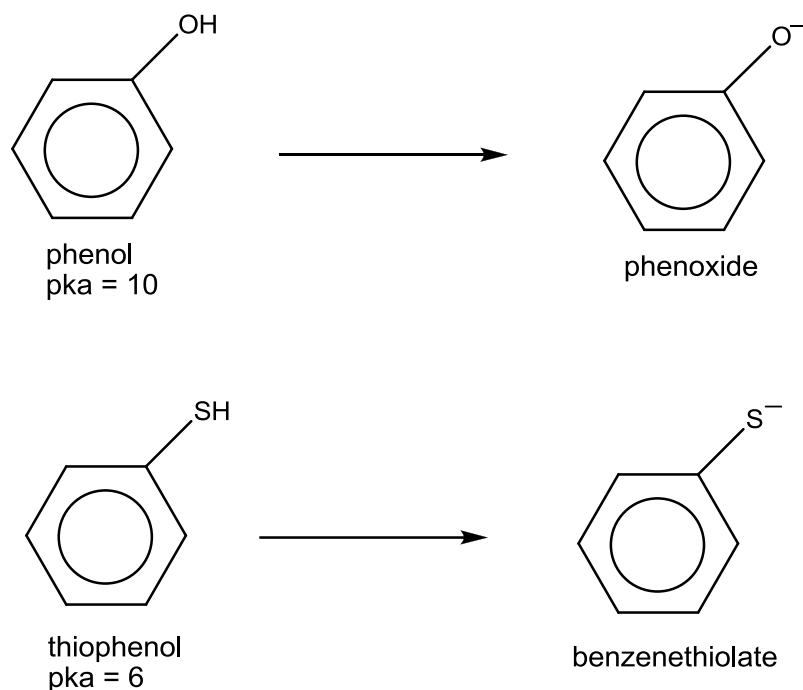
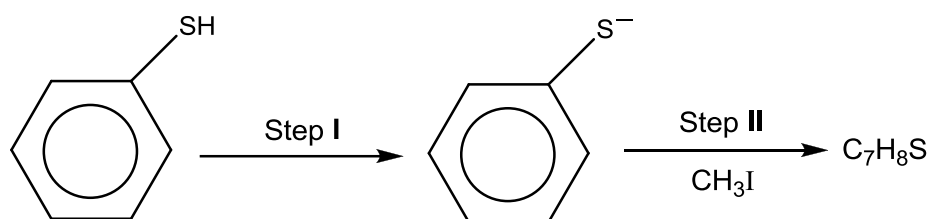


Figure 1

Consider the reaction scheme below.



- (i) Suggest reagent and condition for step I.

[1]

Reagents and Conditions: Na metal at room temperature

OR

Reagents and Conditions: NaOH (aq) at room temperature

- (ii) What *type of reaction* is step I and step II?

[2]

Step I

Acid- Metal (if student answer is Na metal)

Accept: Redox

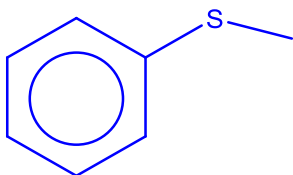
OR

Acid-base if student answer is NaOH(aq)

Step II

Nucleophilic substitution

- (iii) Suggest the structure of C₇H₈S.



[Total: 23]

- 3 Halogenated organic compounds are widespread throughout nature and have a vast array of uses in modern industrial processes. They find many uses in the industries such as solvents and pesticides.

The rate of hydrolysis of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ with aqueous NaOH was studied in a series of 3 experiments in which the initial rate of the reaction was measured. The following results were obtained.

Experiment	$[\text{OH}^-]$ / mol dm^{-3}	$[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}]$ / mol dm^{-3}	Initial Rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.100	0.010	4×10^{-6}
2	0.200	0.010	8×10^{-6}
3	0.200	0.020	16×10^{-6}

- (a) Explain the meaning of the following terms.

(i) order of reaction

(ii) half-life

The order of reaction with respect to a given reactant is the power to which the concentration of that reactant is raised to in the rate equation.

The half-life ($t_{1/2}$) of a reaction is the time taken for the concentration of a reactant to decrease to half its initial value.

[2]

- (iii) Using the data given, derive the rate equation for the hydrolysis of $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$.

For $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$

(inspection method)

Using experiment 1 and 2, when $[\text{OH}^-]$ doubles, keeping $[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}]$ constant, rate doubled. Hence order of reaction with respect to $[\text{OH}^-]$ is one.

Using experiment 2 and 3, when $[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}]$ is doubled, keeping $[\text{OH}^-]$ constant, rate doubled. Hence order of reaction with respect to $[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}]$ is one.

(Substitution method)

Let the rate law be:

$$\text{Rate} = k[\text{OH}^-]^a[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}]^b$$

Using experiment 1 and 2

$$\frac{4 \times 10^{-6}}{8 \times 10^{-6}} = \frac{k(0.100)^a(0.010)^b}{k(0.200)^a(0.010)^b}$$

$$\left(\frac{1}{2}\right)^a = \frac{1}{2}$$

$$a = 1$$

Using experiment 2 and 3

$$\frac{8 \times 10^{-6}}{16 \times 10^{-6}} = \frac{k(0.010)^b}{k(0.020)^b}$$

$$\left(\frac{1}{2}\right)^b = \frac{1}{2}$$

$$b = 1$$

$$\text{Rate} = k [\text{OH}^-][\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}]$$

[3]

- (iv) Using your rate equation, determine a value for the rate constant, including units for this reaction.

$$\text{Rate} = k [\text{OH}^-][\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}]$$

$$6 \times 10^{-6} = k(0.100)(0.010)$$

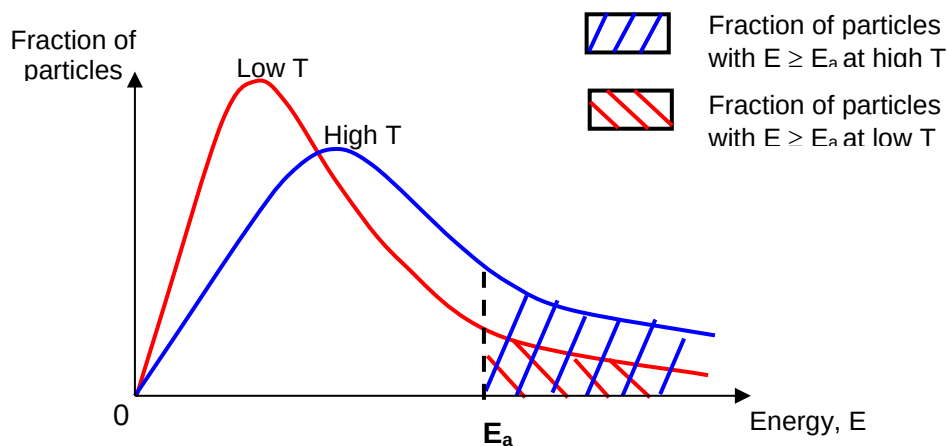
$$k = 6.00 \times 10^{-3} \quad \text{allow ecf}$$

$$k = \frac{\text{Rate}}{[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}][\text{OH}^-]}$$

$$k = \frac{\text{mol dm}^{-3} \text{s}^{-1}}{(\text{mol dm}^{-3})^2} = \text{mol}^{-1} \text{dm}^3 \text{s}^{-1} [1]$$

[1]

- (v) Using the Maxwell-Boltzmann distribution curve, explain how the reaction rate might change with an increase in temperature.



- correct axes
- two graphs with higher temperature graph shifted to the left and lower peak, both graph must start from origin.
- both graphs should be labelled, E_a indicated with shading
- legend of shading provided.

An increase in temperature will result in an increase in the average kinetic energy of the particles.

There is an increase in the fraction of particles with energy equal to or greater than the activation energy, E_a . This is as shown by the larger shaded area at a higher temperature in the above diagram.

This result in an increase in the frequency of effective collisions.

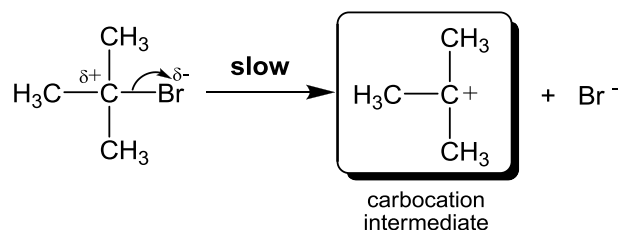
Since rate is proportional to the frequency of effective collision, rate increases.

- (vi) Given that the rate equation for the hydrolysis of $(\text{CH}_3)_3\text{CBr}$ is $\text{rate} = k[(\text{CH}_3)_3\text{CBr}]$, suggest possible mechanisms for **both** of the reactions of $(\text{CH}_3)_3\text{CBr}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ with aqueous NaOH which are consistent with the observed kinetics. [4]

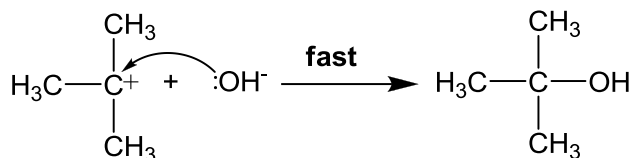
For $(\text{CH}_3)_3\text{CBr}$

Nucleophilic substitution, $\text{S}_\text{N}1$

Step 1 (formation of carbocation):



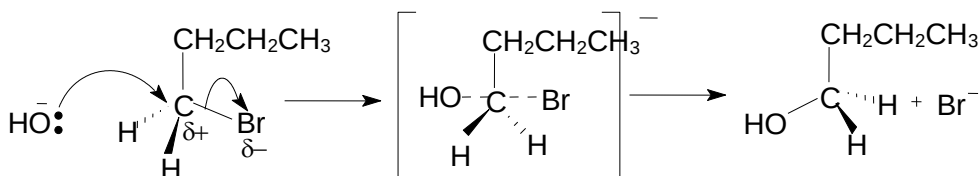
Step 2 (attack by nucleophile)



- δ^+ and δ^- on C-Br
- arrow to show breaking of C-Br bond, arrow must start from the bond
- slow and fast step
- Correct intermediate and Br^- formed
- lone pair on O of OH^-
- arrow starting from lone pair to C^+

For $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$

Nucleophilic substitution, $\text{S}_\text{N}2$



- δ^+ and δ^- on C-Br
- arrow to show breaking of C-Br bond, arrow must start from the bond
- lone pair on O of OH^-
- arrow starting from lone pair to C
- square bracket and negative charge for transition state
- initial reactant and final product should be inversed (solid wedge and dash is not necessary) and Br^- as product

- (b) Bromine reacts with 2-methylpropane in the presence of sunlight to produce mono-brominated products via a free radical mechanism.

- (i) Explain why the mono-bromination of 2-methylpropane results in the formation of two products in unequal amounts.

In the propagation step, two alkyl radicals will be formed. Tertiary $(\text{CH}_3)_3\text{C}^\bullet$ is more stable than primary $(\text{CH}_3)_2\text{CCH}_2^\bullet$. / There are 9 primary H atom and only 1 tertiary H atom.

[1]

- (ii) By quoting appropriate data from the *Data Booklet*, explain the difference in the reactivity of fluorine with 2-methylpropane compare with that of bromine.

Bond energy of $\text{F}_2 = 158 \text{ kJ mol}^{-1}$ Bond energy of $\text{Br}_2 = 193 \text{ kJ mol}^{-1}$

Fluorine will react more vigorously/ more reactive because the F-F bond is weaker than Br-Br bond.

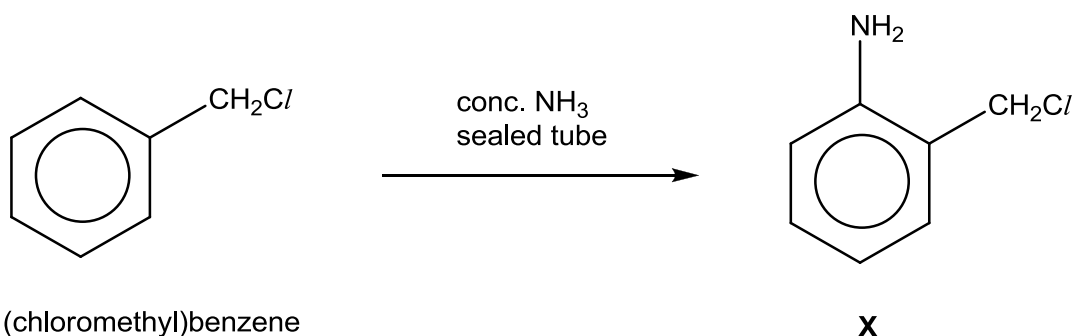
Or

Bond energy of C-F = 485 kJ mol^{-1} Bond Energy of C-Br = 280 kJ mol^{-1}

Fluorine will react more vigorously because C-F bond formed is stronger than C-Br bond.

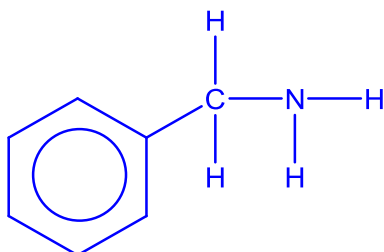
[2]

- (c) A student was not successful in making X from (chloromethyl)benzene with the proposed reaction route as shown.



Instead, a different product with molecular formula $\text{C}_7\text{H}_9\text{N}$ was obtained.

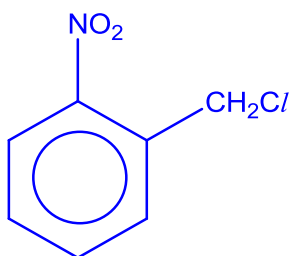
- (i) Draw the displayed formula of the product obtained.



[1]

- (ii) Suggest how **X** can be synthesised from (chloromethyl)benzene. You should include the reagents, conditions and intermediate(s) formed in your answer.

Step 1: Conc HNO₃, Conc H₂SO₄, 30°C < temp < 55°C (30/55 will not be accepted)



Step 2:

- 1) Sn, conc HCl, heat
- 2) aq NaOH, rtp

[3]

[Total: 24]

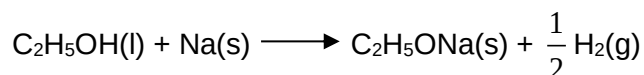
Section B

Answer **one** question from this section

- 4 Hydrogen gas is needed in a large number of processes in the chemical industries such as hydrocracking of petroleum, production of margarine and production of ammonia.

Hydrogen gas can be produced in many chemical reactions.

- (a) One of the reactions that produces hydrogen is the reaction of sodium with ethanol.



This is also a chemical test for the presence of alcohol.

- (i) Write the full electronic configuration of Na.

Electronic configuration of sodium: 1s²2s²2p⁶3s¹

- (ii) Describe how you would use the ionisation energies found in the *Data Booklet* to conclude that sodium is in group 1 of the periodic table.

[1]

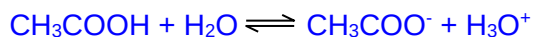
Since there is a large increase between the first ionisation energy (494 kJ mol⁻¹) and the second ionisation energy (4560 kJ mol⁻¹) of sodium. This means that the second electron is removed from the inner principal quantum shell, hence there is only one valence electron in sodium.

- (iii) Ethanoic acid reacts with sodium in a similar way. However, when ethanol and ethanoic acid is reacted with solid sodium carbonate separately, only ethanoic acid gives off a gas that forms white precipitate with calcium hydroxide.

Explain the difference in reactions of ethanol and ethanoic acid with sodium carbonate.

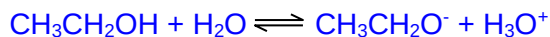
[3]

Reasons: Ethanoic acid is a stronger acid than ethanol hence it is acidic enough to react with sodium carbonate.



For the ethanoate anion, the negative charge can be delocalised over two electronegative oxygen atoms. This disperses the negative charge on the oxygen atom and leads to a stabilisation of the ethanoate anion.

Hence, ethanoic acid is the more acidic.



For the ethoxide anion, the electron-donating ethyl group intensifies the negative charge on the oxygen atom. Thus, the ethoxide anion is less stable.

- (iv) An unknown compound **M**, $\text{C}_8\text{H}_{10}\text{O}$ was found to be an alcohol when tested with sodium metal.

When **M** is treated with hot acidified potassium dichromate, compound **N**, $\text{C}_8\text{H}_8\text{O}$ is formed. **N** gives a yellow precipitate, **P** when warmed with alkaline aqueous iodine.

When **M** is heated with concentrated H_2SO_4 , compound **R**, C_8H_8 is formed. **R** gives compound **S** when reacted with hydrogen chloride gas.

M also gave compound **S** when reacted with phosphorous pentachloride at room temperature.

Deduce the structures of **M**, **N**, **P**, **R** and **S**.

[5]

Explanation not required.

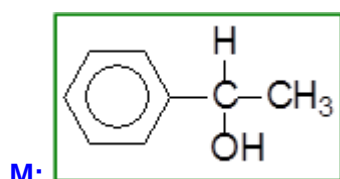
Comparable number of C and H $\rightarrow$ **M** contained benzene ring.

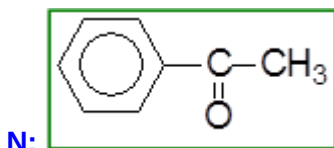
M undergoes oxidation with hot acidified potassium dichromate $\rightarrow$ **N** is aldehyde or ketone

N undergoes oxidation with alkaline aqueous iodine to give yellow precipitate,

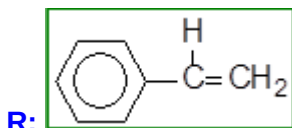
P: CHI_3

$\rightarrow$ **N** has structure $-\text{C}-\text{CH}_3$ and so **M** has $\text{CH}(\text{OH})\text{CH}_3$ structure.



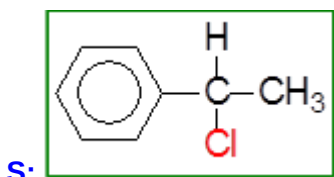


M undergoes dehydration when heated with concentrated $\text{H}_2\text{SO}_4 \rightarrow$ **R** is an alkene

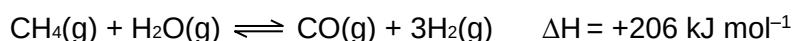


R undergoes electrophilic addition when heated with hydrogen chloride gas $\rightarrow$ **S** is halogenoalkane

M undergoes substitution with phosphorous pentachloride at room temperature $\rightarrow$ **S** is halogenoalkane



- (b) Another reaction that produces hydrogen is the reaction of methane with steam at 1000–1400 K.



This reaction produces large quantities of hydrogen for industrial use.

- (i) State and explain the effect of decreasing the temperature of the reaction on the equilibrium position.

[1]

When temperature of the system is decreased, by Le Chatelier's Principle, the equilibrium will favour exothermic reaction to release heat. Hence, the backward reaction is favoured and equilibrium position shifted left.

- (ii) Deduce the sign of entropy change for the forward reaction.

[1]

Since the no. of moles of gas increases for the forward reaction, disorderness increased as there are more ways to arrange the particles so entropy change (ΔS) is positive.

- (iii) Hence, or otherwise, predict and explain the spontaneity of the forward reaction at low temperature.

[1]

Since $\Delta G = \Delta H - T\Delta S$

When temperature is low, magnitude of $-T\Delta S$ will be smaller than that of magnitude of ΔH . or similar reasoning.

Since ΔS is positive and hence $-T\Delta S$ is negative and ΔH is positive, ΔG would be positive and hence the reaction is not spontaneous

- (iv) Comment and explain if there is any discrepancy in your answers in **b(i)** and **b(iii)**.

[1]

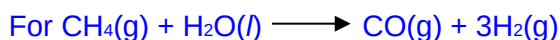
No. Since at low temperature, forward reaction is not spontaneous (ΔG is positive) hence the position of equilibrium position should be on the left/ forward reaction is not favoured

- (v) Calculate the enthalpy change of vaporisation of water of using the data in **Table 1**.

Table 1

Enthalpy change of formation of carbon monoxide	-111 kJ mol^{-1}
Enthalpy change of formation of methane	-75 kJ mol^{-1}
Enthalpy change of formation of water	-285 kJ mol^{-1}

[2]

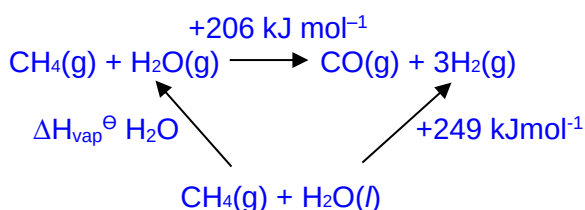


Enthalpy change of reaction

$$= \sum \Delta H_f \text{ product} - \sum \Delta H_f \text{ reactant}$$

$$= (-111 + 3(0)) - (-285 + -75)$$

$$= +249 \text{ kJmol}^{-1}$$

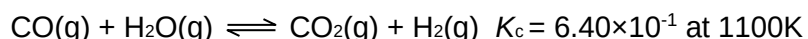


By Hess' law,

$$\begin{aligned}
 \Delta H_{\text{vap}}^\ominus \text{H}_2\text{O} &= +249 \text{ kJmol}^{-1} - (+206 \text{ kJ mol}^{-1}) \\
 &= +43.0 \text{ kJ mol}^{-1}
 \end{aligned}$$

Or any other possible approaches.

- (c) Hydrogen gas can also be produced from the reaction of carbon monoxide with steam.



A mixture containing 0.8 mol of CO and 0.8 mol of H_2O was placed in a 2 dm^3 flask and allowed to come to equilibrium at 1100K.

- (i) Write an expression for an expression for K_c of this reaction.

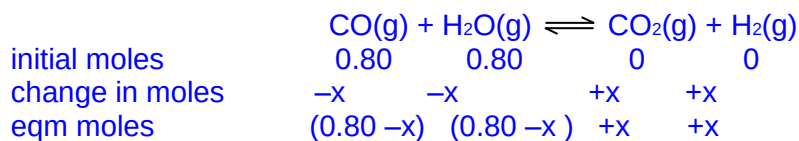
[1]

$$K_c = \frac{[\text{H}_2][\text{CO}_2]}{[\text{CO}][\text{H}_2\text{O}]}$$

- (ii) Calculate the amount of each substance present in the equilibrium mixture at 1100K.

Let the change in number of moles be x .

[2]



$$K_c = \frac{\frac{x^2}{2}}{\left(\frac{0.80 - x}{2}\right)^2} = 6.40 \times 10^{-1}$$

$$0.8 \times 0.8 - 0.8x = x$$

$$x = 0.356$$

At equilibrium

$$n(\text{CO}) = n(\text{H}_2\text{O}) = (0.80 - 0.356) = 0.444 \text{ mol}$$

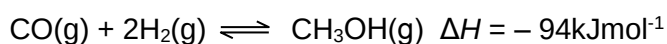
$$n(\text{CO}_2) = n(\text{H}_2) = 0.356 \text{ mol}$$

- (iii) State and explain the effect of decreasing pressure on the equilibrium constant. [1]

Decreasing pressure has no effect on the equilibrium constant as it is only affected by change in temperature.

[Total:20]

- 5 Methanol can be prepared industrially by reacting together carbon monoxide and hydrogen. This is a reversible reaction:



A chemist mixes together 0.114 mol CO(g) and 0.152 mol $\text{H}_2\text{(g)}$ in a container. The container is pressurised and then sealed and the total volume is 200 cm^3 . The mixture is heated to 500 K at constant volume and left to reach equilibrium. The chemist analyses the equilibrium mixture and finds that 0.052 mol CH_3OH has formed.

- (a) Calculate the value of K_c for the equilibrium at 500 K. [2]

	CO(g)	$+$	$2\text{H}_2\text{(g)}$	$\rightleftharpoons$	$\text{CH}_3\text{OH(g)}$
Initial amount/ mol	0.114		0.152		0
Change in amount/ mol	-0.052		-2(0.052)		+0.052
Equilibrium amount / mol	0.062		0.048		0.052
Equilibrium concentration/ mol dm^{-3}	0.31		0.24		0.26

All three equilibrium concentrations

$$K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{H}_2]^2 [\text{CO}]} = \frac{[0.26]}{[0.24]^2 [0.31]}$$

$$= 14.6 \text{ mol}^{-2} \text{ dm}^6$$

- (b) The chemist repeats the experiment using the same initial amounts of CO and H₂. The same procedure is used but the mixture is heated in the 200 cm³ sealed container to a temperature higher than 500 K.
As the gas volume is kept at 200 cm³, the increased temperature also increases the pressure.

Explain why it is difficult to predict how the yield of CH₃OH would change and state what happens to the value of K_c . [2]

At higher pressure, position of equilibrium shifts to the **right** with **fewer gas molecules**.

At higher temperature position of equilibrium shifts to the **left** to favour **endothermic reaction**.

Since **relative effect of pressure and temperature is unknown**, it is difficult to predict how the yield of CH₃OH would change.

The value of K_c **decreases**.

- (c) Methanol can be oxidised to formaldehyde, HCHO, by passing its vapour over copper heated to 300 °C.

Explain why the boiling point of methanol (65 °C) is higher than that of formaldehyde (−19 °C). [2]

Both HCHO and CH₃OH have simple molecular structures.

CH₃OH have stronger hydrogen bonds between the molecules while HCHO have weaker permanent dipole-permanent dipole interactions between the molecules .

More energy is required to overcome the stronger hydrogen bonds between CH₃OH molecules, thus CH₃OH has a higher boiling point.

- (d) State the type of hybridisation shown by the C atom in formaldehyde and draw its hybrid orbitals. [2]

Type of hybridisation: sp^2

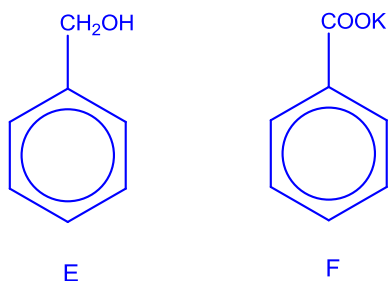


three sp^2 hybrid orbitals with correct shape

- (e) Benzaldehyde, C₆H₅CHO, is the simplest aromatic aldehyde and has a characteristic smell of almonds.

In the presence of a solution of potassium hydroxide, benzaldehyde undergoes disproportionation to give a mixture of two organic products, **E** and **F**. **F** has a higher solubility in water compared to **E**.

- (i) Suggest the structures of **E** and **F**. [2]

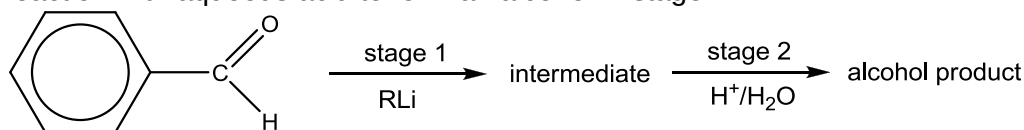


- (ii) Explain why **F** is soluble in water. [1]
 Ionic salt **F** is able to form ion-dipole interactions with water molecules .

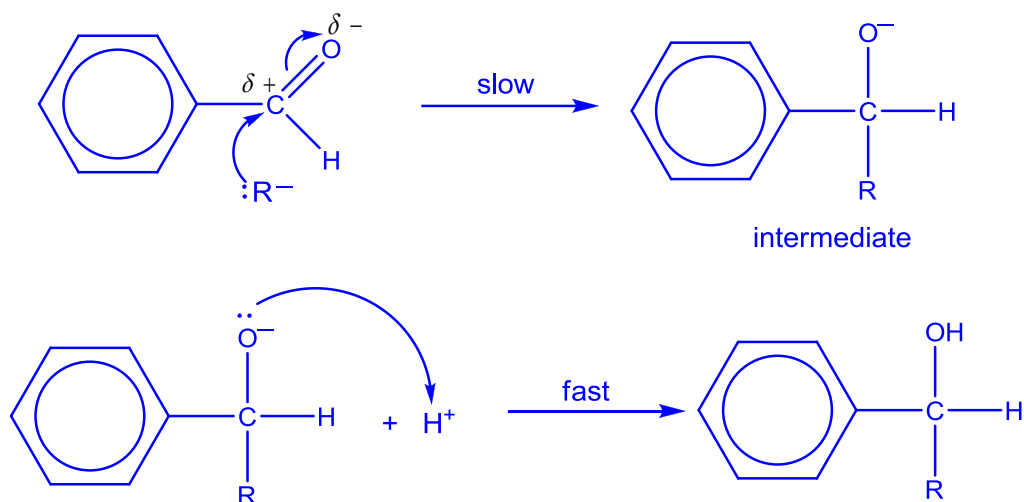
- (f) Alkyl lithium compounds, RLi, can be used to increase the number of carbon atoms in an organic compound. Different alkyl groups, R, add carbon chains with different chain lengths.

RLi provides a source of R^- ions, which act as a nucleophile.

- (i) RLi reacts with benzaldehyde to give an intermediate in stage 1, followed by reaction with aqueous acid to form an alcohol in stage 2.

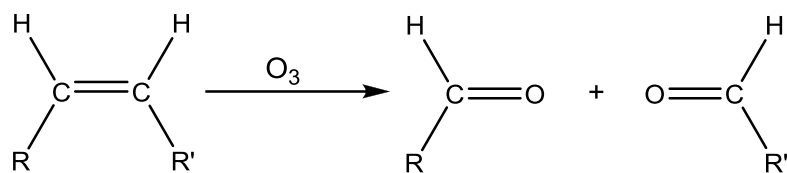


Describe the mechanism for the above reaction, including curly arrows and relevant dipoles. [3]



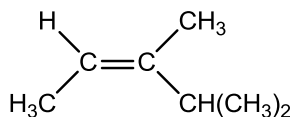
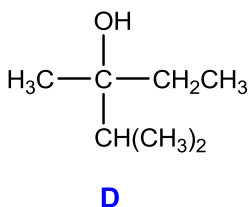
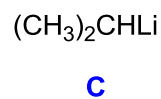
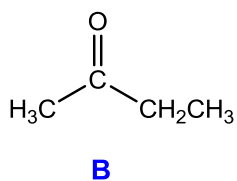
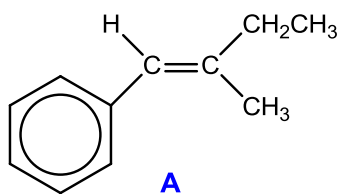
- dipoles on $C=O$ bond & lone pair of electrons on R^- nucleophile
- 2 electron arrows in step 1
- slow and fast step
- correct intermediate
- lone pair of electrons on O^- and electron arrow in step 2
- alcohol product

- (ii) 'Ozonolysis' is a technique used in Organic Chemistry to break open a $C=C$ double bond. During ozonolysis, an alkene reacts with ozone, O_3 . The products are carbonyl compounds as shown below.

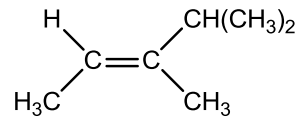


An aromatic alkene, **A** undergoes ozonolysis to give benzaldehyde and **B**, $\text{C}_4\text{H}_8\text{O}$. **B** has no reaction with $[\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$ but gives a yellow precipitate with aqueous alkaline iodine.

B undergoes a reaction with an alkyllithium compound, **C**, followed by reaction with aqueous acid to give an alcohol, **D**, which is optically active. The elimination of H_2O from **D** produces a mixture of four different isomeric alkenes with the formula, C_7H_{14} , only two of which are cis-trans isomers of each other. Suggest the structural formulae of compounds **A** to **D** and the two cis-trans isomers. [6]



Cis and trans isomers



[Total: 20]